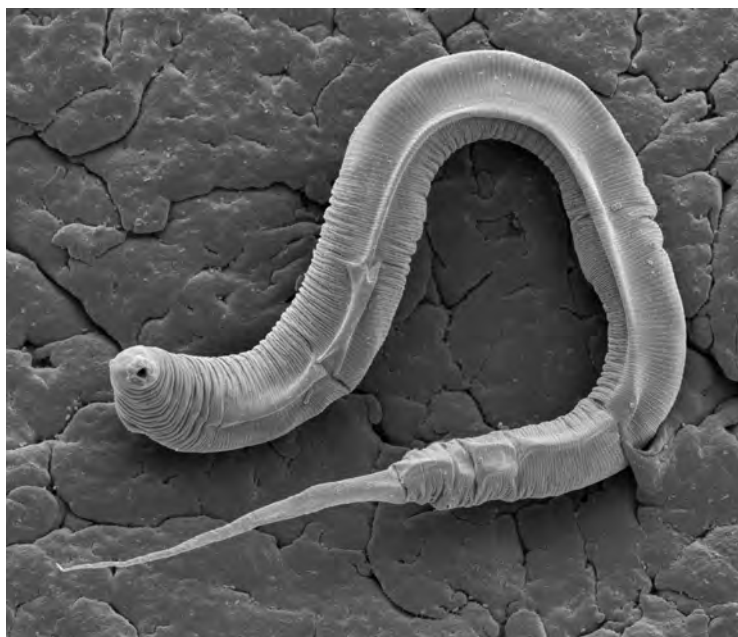


- 3 The photograph shows a roundworm, *C. elegans*.



Magnification $\times 200$

(Source: © STEVE GSCHMEISSNER / SCIENCE PHOTO LIBRARY)

These roundworms are simple animals with a nervous system consisting of 302 neurones.

- (a) The roundworm can respond to stimuli with reflex actions.

A reflex action is a rapid involuntary movement in response to a stimulus that involves receptors.

- (i) Which receptors detect a change in light intensity and stimulate an action potential?

(1)

- ☐ **A** baroreceptors
- ☐ **B** chemoreceptors
- ☐ **C** osmoreceptors
- ☐ **D** photoreceptors

- (ii) A reflex arc carries a nerve impulse from receptors in the skin to the central nervous system in a human.

Describe the structures that a nerve impulse passes through from the skin to the central nervous system.

(3)

- (b) Roundworms respond to physical stimuli by changing the direction in which they swim.

The recovery time of roundworms following habituation was investigated.

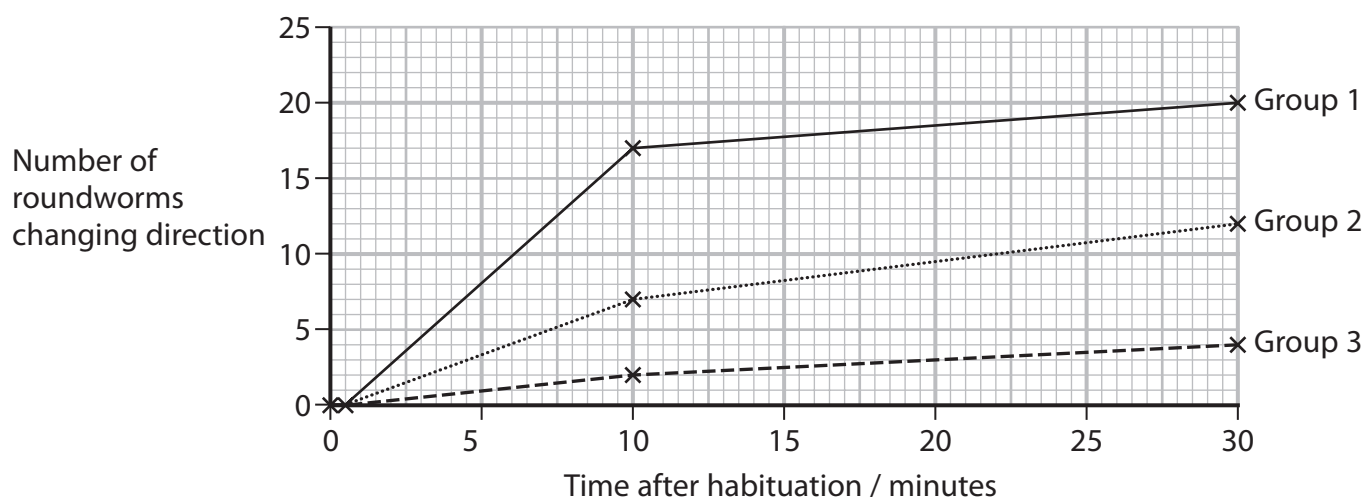
Three groups of 25 roundworms were kept in separate Petri dishes.

The roundworms were habituated to a physical stimulus by tapping the side of the Petri dish at different time intervals.

- Group 1 were habituated with 60 taps at 2-second intervals.
- Group 2 were habituated with 60 taps at 10-second intervals.
- Group 3 were habituated with 60 taps at 60-second intervals.

At 0, 0.5, 10 and 30 minutes after this habituation, each Petri dish was tapped once and the number of roundworms whose swimming direction changed was recorded.

The graph shows the results of this investigation.



- (i) Calculate the percentage difference between the number of roundworms in group 1 and group 2 that changed direction 30 minutes after habituation.

Use the equation:

$$\text{percentage difference} = \frac{\text{difference between two values}}{\text{mean of the two values}} \times 100$$

(2)

Answer %

(ii) Describe **three** conclusions about recovery times after habituation.

(3)

(Total for Question 3 = 9 marks)
